

Report on the PhD dissertation of Bram Vancraeynest-De Cuiper

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Report

The dissertation studies generalized symmetries in quantum lattice models, including their gauging and anomalies, from the perspective of tensor networks and quantum information. Part I is an introduction that reviews matrix product states and the classification of 1+1d gapped phases, fusion categories, and string-net models. Part II consists of lecture notes and five published articles, each preceded by a synopsis and a clear statement of the candidate's own contribution.

The main results are a classification of 1+1d SPT phases protected by frieze (quasi-one-dimensional spatial) symmetries (chapter 6), an explicit constant-depth quantum circuit mapping of Morita-equivalent string-net models onto each other (chapter 7), an extension of the gauging map to fusion category symmetries, together with the result that these gauging maps coincide with the MPO duality intertwiners up to a constant-depth unitary circuit (chapter 8), a construction of stabilizer codes in one dimension higher by iteratively gauging an abelian symmetry and its dual (chapter 9), and twisted gauging of 2+1d abelian lattice gauge theories (chapter 10).

The thesis sits at the interface of tensor networks, topological order, and recent developments in generalized and non-invertible symmetries within the condensed matter and high-energy communities. The most substantial result is, in my view, chapter 8, which provides a concrete lattice counterpart of the algebraic description of gauging by algebra objects known from 2d quantum field theory and complements the gauging of non-invertible symmetries on tensor product Hilbert spaces in the high-energy literature. The results are original and technically correct to the best of my knowledge.

I have the following question for the meeting of the Examination Board:

- What is the physics of the self-dual 2+1d models of chapter 10? Are they gapped, a transition between the toric code and double semion phases, or a stable phase of the extended symmetry?

I have no mandatory modifications to suggest.